Control rack in connection with the tooth quadrant present in barrel unit is used. By moving the control rack front or back the helix position in the plunger meeting the spill port is varied. When the plunger is moved up, the inlet and spill port are cut and hence the fuel is compressed inside the barrel and high pressure is developed in the pump. When sufficient pressure is attained the delivery valve opens and the pressurised fuel present in the pump is supplied to the respective injector as per the firing order and injected inside the combustion chamber.

9.9.2 Distributor Pumps

Use of individual pump and injector in individual cylinders of a multi cylinder engines leads to energy loss (due to the operation of the individual plungers in the pump) and more maintenance in diesel engines. Hence instead of the in line pump, a single unit arrangement can be used which could pressurise the fuel and supply it to the appropriate cylinders at the appropriate time. Such pump is called as the distributor pump. The power for the distributor pump is obtained from the gear present in the cam shaft.

Construction: The distributor pump has a sleeve, rotor, plunger and a delivery unit. Rotor is placed inside the sleeve to rotate and move up and down. For the diesel fuel to enter and leave ports are made in the sleeve. The port is called as the metering ports. The plunger unit is placed inside the rotor unit to operate. There is a gear placed at the bottom of the rotor. The delivery unit of the pump is kept at the top of the rotor. The view of the distributor pump can be seen in Fig 9.9.2.

Operation: Unlike the in line pump, the distributor pump is designed as a single unit in such a way to distribute the diesel fuel to all cylinders. In the rotor a long path is present to pass the fuel. This path is provided to supply the fuel to the individual cylinders depending on the rotation of the suction port. When the engine is started due to the rotors rotation and up and down movement the fuel entering inside the pump is compressed and hence the pressure of the fuel is raised. The pressurised fuel then reaches the delivery valve. From the delivery valve through the high pressure line fuel is supplied to the injectors as per the firing order and sprayed at high pressure. The fuel delivered from the pump is controlled by using a metering rod and the governor unit for controlling the engine's speed. To control and meter the fuel the metering rod and the governor unit are used. The governor unit is connected with the accelerator pedal of the vehicle.

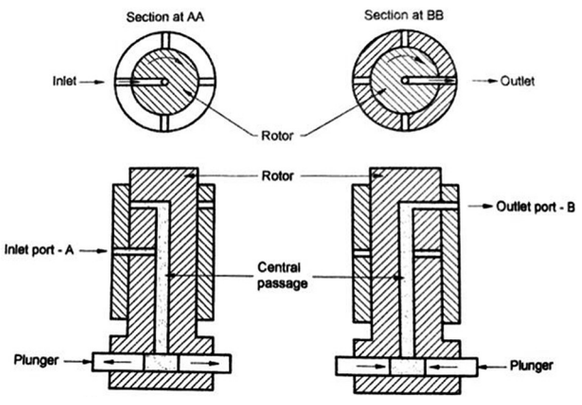


Fig.9.9.2 A view of the Distributor pump

9.10 GOVERNOR

Normally in petrol engines the carburettor unit mixes the air and fuel at correct proportions and supplies to the engine depending upon the engine's requirement. However, in diesel engine air and fuel are

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